

Vernier: calibration-aware acquisition design for IVIM diffusion MRI

Do b-value schemes differ in post-conformal uncertainty calibration at matched scan-time and matched Cramér–Rao precision?

(author list)

Draft — feasibility build

Scope of evidence. The feasibility result (§4.1) is *publication-independent*: it runs entirely on the open, synthetic, MIT-licensed Caliper toolkit and depends on no paper in review. Claims marked ^P are *provisional*: the efficiency-frontier decision values (§4.2) use the decision lens of **Minos**, and the honest-scope statement uses the identifiability-wall result of **Gauge**; both companions are under review. If either changes in revision, the ^P numbers are invalidated and must be re-run (`./ASSUMPTIONS.md`). Every number is regenerated from seeded results by `consistency.py`.

Abstract

Diffusion-MRI acquisition design optimises either the *variance* of the point estimate (the Cramér–Rao bound, CRLB) or the *expected information* about the parameters (Bayesian experimental design, BED / EIG). Neither asks whether the resulting *uncertainty* is *trustworthy*: whether, after a distribution-free conformal correction, the reported error bars are well *calibrated*. We pose and test that question for intravoxel incoherent motion (IVIM) imaging, where the pseudo-diffusion coefficient D^* is notoriously ill-determined. The question is nontrivial because split-conformal restores *marginal* coverage to nominal for every acquisition by construction; what it does *not* equalise—conditional coverage and interval sharpness—is the test. On matched-scan-time, matched-CRLB(D^*) b-value schemes that differ only in how they spend a fixed budget, the post-conformal D^* interval width spans a factor of 393/289 (width-spread $\Delta_{\text{sharp}} = 0.328$, 95% CI [0.200, 0.399]) and the high- D^* conditional coverage spans 0.842–0.901 ($\Delta_{\text{cond}} = 0.059$, CI [0.036, 0.098]), even though marginal coverage is held at ≈ 0.908 for all of them. Both gaps clear pre-registered thresholds with bootstrap CIs excluding zero: **acquisition design moves the part of calibration conformal does not fix, at matched precision**. We are deliberately narrow: Vernier does *not* claim to improve identifiability—a companion analysis^P shows the high- D^* wall is acquisition-robust (CRLB-optimal design moved CRLB(D^*)/tercile-width only $1.25 \rightarrow 1.05$). The claim lives on the calibration-and-decision axis.

I Introduction

The choice of diffusion weightings (b-values) in an IVIM acquisition is an experiment-design problem. Two objectives dominate the literature. The first is *variance-optimal* design: place b-values to minimise the CRLB of the fitted parameters [1, 2, 3]. The second is *information-gain* design: maximise the expected information gain (EIG) about the parameters under a Bayesian experimental-design (BED) framework [4, 5, 6]. Both optimise a property of the *point estimate*—its variance, or the information the data carry about it.

Neither asks the question a clinician actually relies on: when the method reports an error bar, can you believe it, and how much is that error bar *worth* per minute of scan time? Trustworthiness of an interval is *calibration*: does a nominal 90% interval cover the truth 90% of the time, marginally and within clinically relevant strata? Distribution-free conformal prediction [8, 9, 7] restores marginal coverage for essentially any base estimator. This raises a sharp question that, to our knowledge, the acquisition-design literature has not posed:

*At matched scan-time and matched CRLB precision, do different b-value schemes yield differently-**calibrated** uncertainty **after** conformal correction—and therefore different decision value per scan-minute?*

This is not answered by either canon. Marginal coverage is equalised by conformal correction by construction, so it cannot distinguish schemes; and the CRLB / EIG objectives never look at calibration at all. We test the question directly. Because the answer was genuinely unknown to us in advance—and because a negative answer would mean Vernier is not a standalone contribution but a section of the decision-calibration companion—we treated it as a pre-registered *feasibility gate* (§4.1), with thresholds fixed before running and no tuning toward a positive outcome.

2 Honest scope: calibration, not identifiability

We must separate two claims that are easy to conflate. *Identifiability* asks whether the data determine the parameter at all; for IVIM’s D^* the answer is largely no, and—crucially—this does not improve with acquisition design. A companion conformal-coverage analysis^P swept clinical, CRLB-optimal, and dense b-schemes and found the high- D^* identifiability wall *acquisition-robust*: high- D^* coverage barely moved ($0.841 \rightarrow 0.844$) and the ratio $\text{CRLB}(D^*)/\text{tercile-width} \geq 1.05$ for every scheme, so CRLB-optimal design lowered the bound ($1.25 \rightarrow 1.05$) without removing the wall. We take that wall as given.

Vernier therefore makes *no* identifiability claim. It asks the orthogonal question of whether acquisition design moves *calibration*—the trustworthiness and sharpness of the error bar after correction—and the downstream decision value. Our own CRLB implementation (§3) reproduces the companion’s ≈ 1.05 – 1.25 ratio as a cross-check, but the feasibility result below does not depend on it.

3 Methods

Cohort and estimator (synthetic, open). We generate synthetic IVIM cohorts with the open Caliper toolkit: true parameters (D, f, D^*) drawn from physiologically plausible priors (D^* log-uniform on $[10, 100] \times 10^{-3} \text{ mm}^2/\text{s}$, the poorly identified axis), clean bi-exponential signals on a chosen b-schedule, and Rician noise at a chosen SNR. The device under test is Caliper’s torch-free *segmented* least-squares estimator, which reports deliberately overconfident, homoscedastic quantiles—the standard model-based-UQ failure mode conformal is meant to repair. No clinical data are used.

Matched scan-time and matched precision. Scan-time is proportional to the number of acquired volumes; we hold it fixed by using 11 b-values for every scheme in the gate. Among such schemes we select a subset whose expected $\text{CRLB}(D^*)$ —computed from the IVIM Fisher information matrix—agree to within $\pm 10\%$ of the candidate-pool median, so neither budget nor variance can explain a difference between them. Matching precision is conservative: it removes precision as a confounder and makes any surviving calibration divergence *harder* to find, not easier.

Conformal correction and the ruler. For each scheme we split the cohort into calibration and test halves, fit Caliper’s split-conformal / conformalised-quantile-regression (CQR) correction [7] on the calibration half, and apply it to the test half. We score the corrected intervals with a model-agnostic calibration ruler: marginal coverage, quantile expected calibration error (ECE), sharpness (mean interval width), and conditional coverage within true- D^* terciles [10, 11].

Pre-registered gate. The decision was fixed before running. Marginal coverage is *not* the metric: CQR restores it to nominal for every scheme by construction (we report it only as a sanity check). The load-bearing quantities are what conformal does not equalise: *sharpness*, summarised by the across-scheme width spread $\Delta_{\text{sharp}} =$

scheme	CRLB(D^*)	raw cov.	post cov.	width	high- D^* cov.
perfusion-lean	20.5	0.325	0.904	300	0.865
tissue-lean	23.2	0.294	0.895	330	0.842
wide-ends	22.2	0.295	0.908	289	0.901
mid-heavy	23.4	0.252	0.903	393	0.849

Table 1: Feasibility gate at SNR 33. All four schemes share scan-time (11 b-values) and CRLB(D^*) (within $\pm 10\%$). Conformal equalises *marginal* coverage (≈ 0.90); the post-conformal *width* and *high- D^* conditional coverage* do not equalise. Numbers trace to `results/feasibility_gate.json`.

($\max - \min$)/median of the post-conformal D^* interval width; and *conditional coverage*, summarised by the range Δ_{cond} of high- D^* -tercile coverage. We declare the question **feasible** (PASS) iff $\Delta_{\text{sharp}} \geq 0.10$ or $\Delta_{\text{cond}} \geq 0.05$, and that gap’s 95% bootstrap CI excludes zero. The bootstrap is *paired*: schemes share the same true parameters and noise (only the b-scheme differs), and each resample draws both the calibration indices (which set each scheme’s conformal offset, and hence its width) and the test indices (which set coverage).

4 Results

4.1 The feasibility gate: schemes diverge in post-conformal calibration

Table 1 reports 4 matched-scan-time, matched-CRLB(D^*) schemes ($\text{CRLB}(D^*) \in [20.5, 23.4]$), at SNR 33, over 8000 voxels with a 2000-iteration paired bootstrap. CQR restores the marginal D^* coverage of every scheme from a raw 0.252–0.325 to 0.895–0.908 ($\approx$ nominal 0.90). Yet the post-conformal D^* interval *width* ranges from 289 to 393 ($\Delta_{\text{sharp}} = 0.328$, 95% CI [0.200, 0.399]), and the *high- D^* conditional coverage* ranges from 0.842 to 0.901 ($\Delta_{\text{cond}} = 0.059$, CI [0.036, 0.098]). Both gaps clear their pre-registered thresholds with CIs excluding zero: the verdict is **PASS**.

The effect is robust to SNR. At SNR 25, $\Delta_{\text{sharp}} = 0.311$ and $\Delta_{\text{cond}} = 0.056$; at SNR 50, $\Delta_{\text{sharp}} = 0.038$ (the width gap shrinks as the estimator’s D^* error converges) while $\Delta_{\text{cond}} = 0.072$ persists. The conditional-coverage divergence is thus the robust load-bearing finding: marginal conformal correction hides scheme-dependent conditional miscoverage that the acquisition determines.

4.2 Efficiency frontier: decision value saturates with scan-time

We next score, through the decision lens of the companion decision-calibration work^P [12], each scheme’s conformal-corrected D^* posterior on a stylised treat/spare/escalate decision with asymmetric cost. Sweeping scan-time from 7 to 22 b-values, the corrected D^* interval sharpens (316 $\rightarrow$ 198) and the achieved decision utility improves, but with steeply *diminishing returns*: the marginal decision utility per added scan-minute falls 9.9 $\rightarrow$ 2.9 $\rightarrow$ 0.47 across the steps. Crucially, *no* protocol beats the no-scan prior baseline for D^* (every mean utility lies below the prior’s -14.6)^P: acquisition does not rescue D^* *decisions*, consistent with the identifiability wall (§2). The protocol implication is that the clinical 11-b acquisition captures most of the recoverable calibrated decision value; beyond it, additional scan-minutes buy very little, and none of it closes the D^* decision gap.

5 Discussion

The feasibility result establishes a third axis for diffusion-MRI acquisition design, distinct from variance (CRLB) and information (BED/EIG): *post-correction calibration*. At matched precision and matched scan-time, where the variance- and information-optimal objectives are by definition indifferent, schemes still differ substantially in the

sharpness and conditional coverage of their conformally corrected D^* intervals. Because marginal conformal correction equalises only marginal coverage, these differences are invisible to a marginal-coverage audit and to the CRLB, yet they are exactly what a downstream decision sees.

We are deliberately careful about what this does and does not show. **(1)** It is a calibration claim, not an identifiability claim: the companion wall result^P stands, and our own frontier confirms that acquisition does not make D^* decisions beat the prior. **(2)** The divergence *magnitude* is estimator-specific—we measured it on a segmented reference estimator whose realised D^* error, being far from the CRLB, varies with sampling geometry; whether an efficient amortised posterior narrows the gap is the natural next experiment. **(3)** The sharpness gap is SNR-dependent (it shrinks at high SNR), whereas the conditional-coverage gap persists, so the latter is the more durable signal. **(4)** “CI excludes zero” is a weak bar for a non-negative range statistic; the substantive criterion is the pre-registered *magnitude* threshold, which both gaps clear.

6 Limitations and reproducibility

All data are synthetic and PHI-free. The feasibility gate (§4.1) is SOLID: it runs on the open Caliper toolkit alone and depends on no manuscript in review. The decision-value frontier (§4.2) and the honest-scope wall statement are PROVISIONAL—they consume the in-review decision-calibration^P and conformal-coverage^P companions—and are re-validated by one command when those land (`./ASSUMPTIONS.md`, `./PROMOTION.md`). Every number above is emitted by `consistency.py` from the seeded result files (`results/feasibility_gate.json`, `results/efficiency_frontier.json`); the manuscript contains no hand-entered figures.

References

- [1] A. Lemke, B. Stieltjes, L. R. Schad, F. B. Laun. Toward an optimal distribution of b values for intravoxel incoherent motion imaging. *Magn. Reson. Imaging* 29(6):766–776, 2011. doi:10.1016/j.mri.2011.03.004.
- [2] Q. Zhang, Y.-X. Wang, H. T. Ma, J. Yuan. Cramér–Rao bound for IVIM diffusion-weighted imaging fitting. *35th IEEE EMBC*, 511–514, 2013. doi:10.1109/EMBC.2013.6609549.
- [3] O. Jalnefjord, M. Montelius, G. Starck, M. Ljungberg. Optimization of b-value schemes for estimation of the diffusion coefficient and the perfusion fraction with segmented IVIM model fitting. *Magn. Reson. Med.* 82(4):1541–1552, 2019. doi:10.1002/mrm.27826.
- [4] D. C. Alexander. A general framework for experiment design in diffusion MRI and its application in measuring direct tissue-microstructure features. *Magn. Reson. Med.* 60(2):439–448, 2008. doi:10.1002/mrm.21646.
- [5] Y. Dai, X. Jia, T. Aguilera, K. Jiang, A. Perez Rodriguez, I. Vanhaezebrouck, J. Deng. Optimizing IMPULSED acquisition protocols for clinical 3T scanners through Bayesian experimental design. arXiv:2604.06020 [physics.med-ph], 2026. doi:10.48550/arXiv.2604.06020. (Preprint; cited from arXiv metadata.)
- [6] J. Iollo, G. Oudoumanessah, C. Lartizien, M. Dojat, F. Forbes. Active MRI acquisition with diffusion-guided Bayesian experimental design. arXiv:2506.16237 [eess.IV], 2025. doi:10.48550/arXiv.2506.16237.
- [7] Y. Romano, E. Patterson, E. J. Candès. Conformalized quantile regression. *NeurIPS* 32:3538–3548, 2019.
- [8] V. Vovk, A. Gammerman, G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi:10.1007/b106715.
- [9] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman. Distribution-free predictive inference for regression. *J. Amer. Statist. Assoc.* 113(523):1094–1111, 2018. doi:10.1080/01621459.2017.1307116.
- [10] T. Gneiting, A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *J. Amer. Statist. Assoc.* 102(477):359–378, 2007. doi:10.1198/016214506000001437.

- [11] V. Kuleshov, N. Fenner, S. Ermon. Accurate uncertainties for deep learning using calibrated regression. *ICML* 80:2796–2804, 2018.
- [12] Minos: the decision value of a calibrated uncertainty (companion, in review). Provides the decision lens used in §4.2.